AI Painting Development and Applications

HUAWEI CLOUD
EVERYTHING AS A SERVICE

Foreword

- The rapid development of science and technology extends the application of AI in various fields. Notably, AI painting is attracting people's attention as a new art form.
- This course covers the development history, technical principles, and applications of AI painting in product creation.

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Objectives

- Upon completion of this course, you will:
  - Understand the history of AI painting
  - Be familiar with the typical applications of AIGC

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1. **AI Painting Overview**
2. AIGC Transforms the Content Industry
3. AIGC Typical Applications

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AI Painting Background

- AI painting technology has developed largely in parallel with computer vision, deep learning, and other technologies. Computer vision enables computers to understand images and videos. Deep learning enables computers to process visual information more efficiently, working just like the neural network of a human brain.
- Early AI painting applications mainly used traditional computer graphics technologies, such as vector graphics and basic image synthesis.
- Owing to the rapid development of deep learning, especially the breakthroughs in algorithms such as convolutional neural networks (CNNs) and generative adversarial networks (GANs), AI has contributed significantly to artistic creation.

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AI Painting History

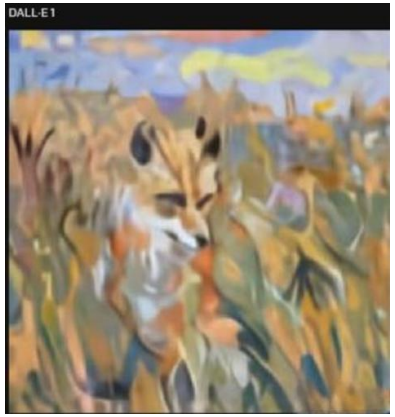
- In the 1950s, computer art pioneers, such as Harold Cohen, began to explore how to generate artworks using computers. He developed a program called "AARON". It could generate simple paintings.

- In August 2022, Stable Diffusion (an open-source AI painting tool) was released. It was easy to deploy and use.

- In 2012, Andrew Ng and Jeff Dean from Google used deep learning to train a model that could generate blurred cat face images based on a large number of cat face images.

- In 2014, Ian Goodfellow proposed GAN, a powerful model that could generate high-quality images.

- At the beginning of 2021, OpenAI released the DALL-E system, which could create content based on input text prompts.



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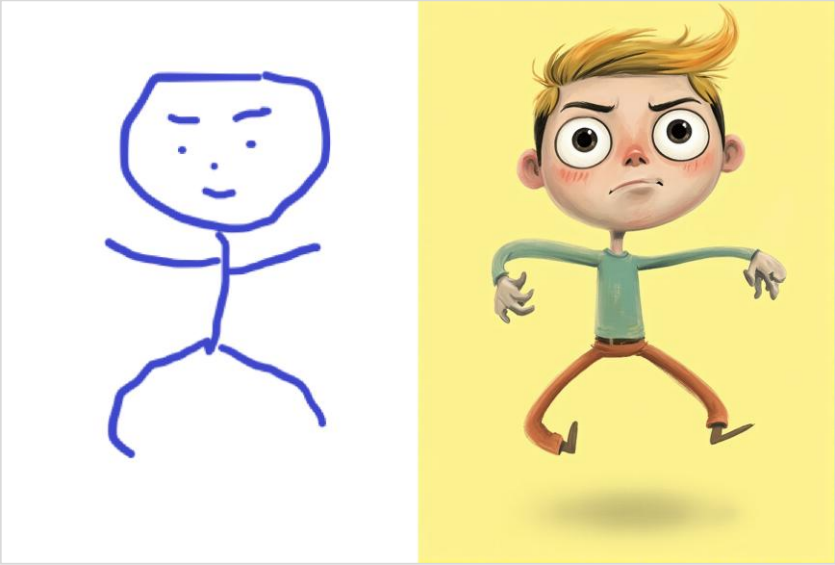
Content Production Mode Transformation

- Content creators: Humans used to create content alone. Now AI provides design ideas and generates materials for humans.
- Creation process: AI combines multiple steps (such as sketch drawing and material collection) from the traditional creation process into one, quickly generates initial images based on input text, and then optimizes the images. AI shortens the creation period significantly.



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Higher Efficiency and Quality Assurance

- Efficient generation: A large number of images can be generated in a short period of time to meet the requirements of different services, such as material production in advertising.
- Quality improvement: Algorithms and models are continuously optimized to improve the quality of generated images in terms of resolution, color matching, and details. Some generated images are even better than those created by humans.

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Typical Applications

- Advertising: AI can quickly generate creative ad images based on product features and target audience. For example, it can generate images that show the functions of a new phone.
- Game development: AI can generate scenes and character images in games. For example, it can quickly design maps of virtual worlds.
- E-commerce: AI can generate product images, such as outfit previews, to attract customers.
- Film and television production: AI can generate storyboards, scenes, and human images to present the ideas of directors and screenwriters.

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Diverse and Innovative Content

- More styles: AI can generate images in diverse artistic styles, including realistic and abstract styles, as well as oil and ink paintings.
- Creativity: AI provides inspiration for creators. For example, AI can generate unique biological images for sci-fi works.
- Personalized content: Based on the prompts input by users, AI can generate content that meets their preferences, such as custom avatars and wallpapers.
- Better experience: AI continuously optimizes generated content based on user feedback, thereby enhancing user engagement and satisfaction.

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Summary

- History of AI painting
- Typical applications of AIGC

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Recommendations

- Huawei Cloud websites
  - Huawei Cloud official website: https://www.huaweicloud.com/intl/en-us/
  - Huawei Cloud Developer Institute: https://edu.huaweicloud.com/intl/en-us/

Huawei Cloud Developer Institute

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